

6.1 INTRODUCTION

Based on service, the microbiome sequencing services market is segmented into sample preparation services, sequencing and library preparation services, and data analysis services. A majority of the service providers/key market players offer all three types of services as part of the microbiome analysis workflow. Variations in sample size, associated pricing, use of different technologies, and the strength of local players are the key factors driving the revenue generation capacity of the above-mentioned market segments. The sequencing and library preparation segment accounted for the largest share of ~48% in 2022, followed by the sample preparation segment. The sequencing and library preparation segment is also estimated to grow at the highest CAGR of 14.7% during the forecast period.

TABLE 13 MICROBIOME SEQUENCING SERVICES MARKET, BY SERVICE, 2021–2028 (USD MILLION)

Service	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Sample Preparation	79.35	90.10	102.25	116.14	132.03	150.63	172.45	198.29	14.2%
Sequencing and Library Preparation	107.34	122.43	139.58	159.26	181.88	208.45	239.73	276.91	14.7%
Data Analysis	33.34	37.64	42.48	47.99	54.24	61.53	70.03	80.04	13.5%
Total	220.03	250.17	284.31	323.39	368.16	420.61	482.21	555.24	14.3%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

6.2 SAMPLE PREPARATION

6.2.1 NEED FOR SPECIFIC SAMPLE PREPARATION PROTOCOLS BASED ON SAMPLE TYPE TO DRIVE GROWTH

This segment includes services that are applied for the collection, storage, and processing of samples subjected to microbiome sequencing. It is imperative to ensure that microbiome analysis is reproducible. Reproducibility is affected by several factors, such as sample size, storage methods, and collection methods. Microbiome sequencing helps in evaluating microbial communities across different applications, including real-time functional profiling (metatranscriptomics), functional profiling (metagenomics), and high-level community profiling (marker gene studies—16S rRNA, ITS, and 18S rRNA). Across all these applications, variation in sample collection, storage, and shipping must be avoided to maintain the reproducibility of the outcome. Sample preparation protocols differ based on sample types (oral microbiomes, gut microbiomes, and skin microbiomes), which has created robust revenue generation opportunities for service providers/players operating in the microbiome sequencing market.

TABLE 14 MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY REGION, 2021–2028 (USD MILLION)

Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
North America	45.42	51.34	58.01	65.60	74.25	84.33	96.11	110.02	13.7%
Europe	24.57	27.99	31.87	36.32	41.42	47.40	54.44	62.80	14.5%
Asia Pacific	6.76	7.90	9.22	10.76	12.56	14.71	17.27	20.35	17.2%
Latin America	1.54	1.70	1.88	2.08	2.30	2.55	2.83	3.16	10.9%
Middle East and Africa	1.06	1.16	1.27	1.38	1.51	1.65	1.80	1.97	9.2%
Total	79.35	90.10	102.25	116.14	132.03	150.63	172.45	198.29	14.2%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 15 NORTH AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
US	41.16	46.57	52.66	59.59	67.50	76.73	87.52	100.25	13.7%
Canada	4.25	4.78	5.35	6.01	6.75	7.60	8.60	9.76	12.8%
Total	45.42	51.34	58.01	65.60	74.25	84.33	96.11	110.02	13.7%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 16 EUROPE: MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Germany	5.32	6.08	6.95	7.94	9.08	10.42	12.01	13.89	14.9%
UK	4.76	5.46	6.27	7.20	8.27	9.54	11.03	12.82	15.4%
France	3.76	4.28	4.88	5.57	6.35	7.27	8.36	9.65	14.6%
Italy	1.06	1.20	1.37	1.55	1.77	2.02	2.31	2.66	14.3%
Spain	0.81	0.92	1.05	1.19	1.35	1.53	1.75	2.01	14.0%
Rest of Europe	8.86	10.04	11.36	12.87	14.60	16.62	18.97	21.76	13.9%
Total	24.57	27.99	31.87	36.32	41.42	47.40	54.44	62.80	14.5%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 17 ASIA PACIFIC: MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
China	2.86	3.34	3.90	4.56	5.32	6.23	7.32	8.63	17.2%
Japan	1.33	1.56	1.84	2.15	2.53	2.98	3.51	4.16	17.8%
India	0.17	0.20	0.23	0.27	0.31	0.36	0.43	0.50	16.9%
Rest of Asia Pacific	2.40	2.79	3.25	3.78	4.40	5.13	6.00	7.05	16.8%
Total	6.76	7.90	9.22	10.76	12.56	14.71	17.27	20.35	17.2%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

TABLE 18 LATIN AMERICA: MICROBIOME SEQUENCING SERVICES MARKET FOR SAMPLE PREPARATION, BY COUNTRY, 2021–2028 (USD MILLION)

Country/Region	2021	2022	2023	2024	2025	2026	2027	2028	CAGR (2023–2028)
Brazil	0.72	0.80	0.89	0.98	1.08	1.20	1.34	1.49	11.0%
Rest of Latin America	0.82	0.90	1.00	1.10	1.21	1.35	1.49	1.66	10.8%
Total	1.54	1.70	1.88	2.08	2.30	2.55	2.83	3.16	10.9%

Source: National Human Genome Research Institute (NHGRI), National Center for Biotechnology Information (NCBI), Genomics Research and Development Initiative (GRDI), UK Research and Innovation (UKRI), Microbiome Foundation, Centre for Genomic Regulation (CRG), Human MetaGenome Consortium Japan (HMGJ), Microbiome Research Centre (MRC), Center of Innovation in Personalized Medicine (CIPM), Department of Biotechnology (GENOME India and Microbiome), South African Society for Human Genetics, Food and Drug Administration (FDA), Annual Reports, SEC Filings, Investor Presentations, and MarketsandMarkets Analysis

6.3 SEQUENCING AND LIBRARY PREPARATION

6.3.1 SEQUENCING AND LIBRARY PREPARATION SERVICES TO DOMINATE MARKET DURING FORECAST PERIOD

This segment includes sequencing and library preparation services for samples subjected to microbiome sequencing. This is a crucial step in microbiome analysis, wherein either reference sequences are created for samples or sequencing outputs are compared against the existing sequences. Different approaches are employed in microbiome sequencing and library preparation, including 16S rRNA sequencing, shotgun metagenomics, qPCR testing, and short-chain fatty acids (SCFA) analysis. Service providers use either state-of-the-art equipment and software or purchase commercial systems and software solutions. Illumina, Inc. (US) and PacBio (US) are among the popular brands adopted by a majority of service providers to offer sequencing and library preparation services. The expanding pool of service providers, particularly the ones offering NGS services, is the key contributor to the dominance of this segment.